

Project Documentation: Supply Chain & Logistics Control Tower

Supply Chain Intelligence & Operations BI Portfolio Project Specification

Problem Statement

Modern global supply chains generate massive streams of transactional, inventory, and shipment data across multiple warehouses, freight carriers, and distribution centers. Supply chain leaders often face bottlenecks, delayed shipments, unexpected stockouts, and rising freight costs due to a lack of real-time visibility. The objective of this project is to build an executive Supply Chain & Logistics Control Tower dashboard that provides end-to-end operational visibility, monitors shipment fulfillment rates, and tracks carrier performance and inventory bottlenecks.

Objectives

- Architect a centralized data model integrating warehouse, inventory, and supplier/carrier data feeds.
- Build advanced DAX measures for core logistics performance metrics (On-Time In-Full Rate, Freight Cost Variance, Inventory Turnover).
- Design an interactive control tower dashboard enabling real-time exception monitoring and bottleneck detection.
- Track shipment status, transit times, and regional distribution performance.
- Provide executive-level insights to drive cost reduction and improve SLA compliance across logistics partners.

Tools & Architecture Used

- **Data Modeling:** Star Schema / Relational Snowflake architecture (Shipment Fact, Warehouse Dimension, Carrier Dimension)
- **Data Transformation & Cleaning:** Power Query / SQL ETL pipelines
- **Calculations & Modeling:** DAX (Data Analysis Expressions for OTIF and cost metrics)
- **Visualization & BI:** Power BI Desktop (Control Tower Executive UI)

Key Performance Indicators (KPIs)

- **OTIF Rate (On-Time In-Full):** Percentage of orders delivered on time and complete without damages.
- **Total Freight Cost:** Cumulative shipping and logistics expenditure across carriers.
- **Average Transit Time:** Mean duration from warehouse dispatch to customer delivery.
- **Inventory Stockout Rate:** Frequency of stock depletion across distribution hubs.

Visualizations Created

- **Executive Control Tower Summary:** Top-level KPI summary cards for OTIF, Freight Spend, and Transit Time.
- **Shipment Status Breakdown:** Donut chart tracking in-transit, delivered, delayed, and canceled shipments.

- **Carrier Performance Analysis:** Clustered bar chart comparing cost vs. on-time delivery rate by logistics carrier.
- **Warehouse & Regional Fulfillment:** Geospatial or column chart analyzing throughput across fulfillment centers.

Expected Outcome

The Control Tower dashboard provides logistics managers with real-time operational transparency, enabling proactive exception handling, optimized carrier contract negotiations, and significantly improved supply chain reliability.